

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1501

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Abhishesh Reddy (192525462) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abhishesh Reddy

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

evaluate the effectiveness of distributed computing models in processing large scale scientific research data, and justify how parallel processing improves computations efficiency and scalability compared to centralized system.

Ans: Scientific research applications such as climate modelling, astronomy, genomics, and particle physics and generate extremely large amounts of data. processing this data using single centralized computer can take a long time and may have limited memory and processing capacity.

* Distributed computing divides a part of the research smaller tasks and process them simultaneously on multiple computers.

Technologies such as Hadoop MapReduce, Apache Spark, distributed clusters and cloud computing can be used

* Advantages of Parallel Processing:-

* multiple processes execute different parts of the problem simultaneously

* large datasets can be divided across multiple machines

* Process time significantly reduced.

Conclusion:- Distributed computing is highly effective for specific research because it enables researchers to process massive datasets faster and expanded computational resources when required.

Q1

Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer.

Ans

A multinational retail company collects customer data from websites, mobile applications, physical stores, payment systems and social media. When this data is stored separately, the company cannot obtain a complete view of customer behaviour.

* A unified big data architecture can be designed as follows:

- * collect online and offline data using Apache NiFi, API
- * Store structured and unstructured data in Data Lake HDFS, Amazon S3 or Azure Data Lake.
- * Process real time customer activity using Apache Spark Streaming or Apache Flink
- * Store frequently accessed customer information in NoSQL database such as MongoDB, Cassandra, or HBase
- * Apply ML algorithms to analyze purchase history, browsing patterns, locations and preferences
- * Use recommendation models such as collaborative filtering and content-based filtering

Result:- The company obtains a 360-degree customer view, provides personalized product recommendations, improves marketing campaigns and increases customer satisfaction and sales.

Design a Fraud detection Framework for a digital Payment Platform using big data analysis, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

A digital Payment Platform processes thousands or millions of transactions continuously. Fraud Detection must therefore operate in real time.

A Suitable Framework includes

- * Transaction Data is collected through Apache Kafka
- * Real time processing is performed using Apache Spark Streaming or Flink
- * Historical transaction data is stored in HDFS, cloud storage or NoSQL Databases.
- * Important features such as transaction amount, location, device, transaction, frequency IP, address, and login behaviour are analyzed.
- * ML algorithms such as Random Forest, Logistic Regression, XGBoost and Neural Networks can calculate a fraud & risk score

* Result:- The Framework provides fraudulent and legitimate transactions. They can identify unusual customer behaviour and previously unseen financial losses improves detection accuracy, and protects customers.

Q.4

Assess the challenges of storing and processing multimedia data such as video, audio and images in a big data environment and recommend suitable technologies storage mechanisms for efficient management

Ans: Multimedia data includes images, audio, files, video files, CCTV recordings, and medical images. Such data creates challenges because multimedia files are generally large and unstructured.

Major challenges include:-

- * Very high storage requirements,
- * large Network Bandwidth requirements.
- * Different formats and compression standards
- * Difficulty in searching unstructured files
- * High processing requirements for video and image analysis
- * Back up and replication costs

Suitable technologies include:-

- * HDFS for distributed multimedia storage
- * Amazon - S3 - Azure, Blob storage, or Google S for scalable object storage
- * mongo DB or cassandra for metadata processing
- * Gpus and deep-learn frameworks for image and video analysis

conclusion:- Object storage combined with distributed processing provides a scalable and cost-effective solution.

Cluster recovery and business continuity strategy for a cloud based big data platform, how redundancy, replication, and automated failover mechanism ensure operational resilience

Cloud computing - Based Big data platform must continue to work even when servers, networks, storage systems or entire data centres fail.

A cluster recovery strategy should include

- * maintain multiple copies and important data through replication
- * store backups in geographically different cloud regions.
- * use automated and scheduled backup and restore mechanisms
- * configure active - primary and standby - secondary systems
- * use load balancers to distribute traffic
- * Implement health monitoring for servers and services

Redundancy:- prevents dependence on one server or storage device

- * Automated Failover:- redirects user and work loads to a healthy server or region when failure occurs

Q6) Analyze the impact of data quality issues such as inconsistency, duplication and missing values in a large scale analytical platform, and propose data governance strategies to improve analytical reliability

Ans:- Poor quality data can produce incorrect analytical result and poor business decisions.

Common data quality problems include

- * Inconsistency:- The same information appears different in different systems

- * Duplications:- The same customer or transaction appears several times

- * missing values:- Important attributes are absent

- * Incorrect or outdated information

- * Different data formats and standards

- * Establishes common data standards

- * Perform data validation during ingestion

- * Use ETL processors for data cleaning

- * Apply duplicates and data matching

- * maintain a master data management system

- * Good data governance improves accuracy, consistency, reliability, security and trust in analytics

design a scalable recommendation system for an online streaming service using big data technologies and justify how ever analytics and distributed processing enhance recommendations and accuracy and user engagement.

Streaming platform must recommend movies, songs or videos according to individual user the interests

The system collects information such as:-

- * Viewing history
- * Searching history
- * Likes and ratings
- * watch duration
- * Skipped content
- * Device information
- * Genre preferences

* User activities can be streamed through Apache Kafka and processed using Apache Spark large amounts of in HDFS

* Collaborative filtering - recommends content liked by

Similar user

* Content - Based filtering - recommends content similar

Result:- The system improves recommendation accuracy viewing time, user engagement, customer satisfaction and retention.

evaluate the role of cloud computing in modern data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility and security.

cloud computing plays an important role in modern Big data system because it provides storage and computing resources through the internet. Major advantages include:-

- * Scalability :- Resources can be increased or decreased according to workload
- * Cost efficiency :- organizations avoid purchasing expensive physical infrastructure
- * Flexibility :- Different computing, database, AI and analytics services can be accessed easily
- * Availability :- Cloud platform provide replication and disaster recovery facilities.
- * Rapid deployment :- Big data clusters can be created quickly
- * Global accessibility :- user can access services, from the different geographical locations

Conclusion:- cloud computing provides excellent scalability and flexibility and for big data applications.

typically analyze ethical and legal challenges, associated with big data collections and analytics. and recommends organizational policies that ensure responsible data usage, regulatory compliance and user trust

Give me an

Ans:- Big data analytics creates important ethical and legal issues because organizations collect large amounts of personal and behaviour information.

major challenges include

- * collection of data without proper consent
- * privacy violations
 - * unauthorized sharing of personal information
 - * Data Breaches and cyberattacks
- * Algorithms Bias and discrimination.
- * Excessive surveillance.
- * lack of transparency about how data is used.
- * violation of data protection laws.

Conclusion:- Responsible big data management requires privacy, transparency, security, fairness, accountability and legal compliance. These practices improve customer trust and reduce legal and reputational risks.

Q1 Develop a real-time analytics framework for transportation and logistics management, and justify and how big data technologies improve route optimization, fuel efficiency and operational decision-making.

Ans: Transportation companies continuously generate GPS, vehicle traffic, weather, fuel and delivery information.

A real-time analytics framework can include

- * GPS and IoT sensors collect vehicle information.
- * Apache Kafka receives continuous data streams.
- * Spark Streaming or Apache Flink processes the data in real time.
- * It improves fuel efficiency by selecting shorter routes, reducing idle time, and monitoring driving behaviour.
- * Real-time dashboards help manager responds as immediately to vehicle delays, breakdowns, traffic congestion and changes in customer demand.

Results:- The framework reduces transportation costs, improves delivery performance, saves fuel, and supports faster operational decisions.